AKENTEN APPIAH-MENKA UNIVERSITY
of Skills Training *and* Entrepreneurial Development

Hello!!!

Silence Hour

Use this time to revise your notes before we begin.

AKENTEN APPIAH-MENKA UNIVERSITY
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ICT233

Web Technologies and Design

Lecture Six

Introduction to CSS

1.0 Overview of CSS

2.0 CSS Document Structure

3.0 CSS Selectors and Declarations

4.0 Applying CSS to HTML documents

Lecture Objectives

- By the end of the lecture, the student will be able to:
  - Explain CSS
  - Describe the CSS document structure.
  - Describe how CSS rules are applied to web pages
  - Write CSS rules and apply them to HTML pages

What is a CSS?

- CSS stands for Cascading Style Sheets.
- CSS is a language that defines style constructs such as fonts, colors and positions which are used to describe how information on a web page is formatted and displayed.
- CSS is a style language that defines layout and appearance of HTML documents.
- *CSS*, is the recommended way to control the presentation layer in web development projects by creating robust style rules.

Overview of CSS - 1

- A style sheet is a grouping of formatting instructions that can control the appearance of many HTML pages at once.
- CSS enables you to set a great number of formatting characteristics that were never possible before with any amount of effort.

Benefits of Using CSS

- By externalizing the presentation layer, CSS offers a number of significant benefits:
  1. All styling is kept in a limited number of style sheets.
  2. There is some saving on bandwidth.
  3. The separation of content from presentation makes it easier for site owners to reuse the content for other purposes.
  4. Separate styling rules can be used for different output media.

Types of Style Sheets

- Style sheets are classified according to how they are applied to the HTML documents.
- There are three types of style sheets.
- These are:
  - External Styles
  - Internal (Embedded) Styles.
  - Inline Styles

External Style Sheet

- An external style sheet is ideal when the style is applicable to many pages.
- With an external style sheet, you can change the look of an entire web site by changing one file.
- Each page must reference the external style sheet using the **<link>** tag.
- The **<link>** tag goes inside the **<head>** section of your HTML page as shown in below:

```
<head>
    <link rel="stylesheet" type="text/css" href="mystyle.css“ />
</head>
```

Internal Style Sheet

- An internal style sheet can be used if one single document has a unique style.
- Internal styles are defined in the <head> section of an HTML page, by using the **<style>** tag, like this:
-

```
<head>
<style type="text/css">
    body {background-color:yellow}
    p {color:blue}
</style>
</head>
```

Inline Styles

- An inline style can be used if a unique style is to be applied to one single occurrence of an element on a web page.
- To use inline styles, use the style attribute in the start tag of the relevant element.
- The style attribute can contain any CSS property value pairs.
- The example below shows how to change the text color and the left margin of a paragraph:

```
<p style="color:blue;margin-left:20px;">This is a paragraph.</p>
```

Using the and Elements

- The ** ** (division) and the **** (span of words) elements are designed to be used with CSS.
- These elements let you specify logical divisions within a document that have their own names and style properties.
- ** ** is a block-level element and **** is its inline equivalent.

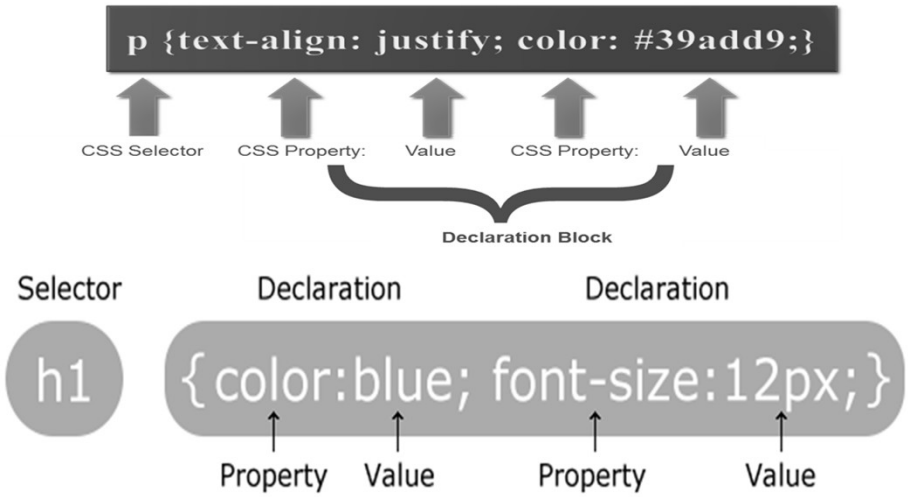
CSS Document Structure

- In order to work in CSS, one must be familiar with some markup languages like HTML or XHTML so that you can add personalized styles on the markup documents.
- This is because CSS rules are applied to HTML or XHTML elements.

CSS Rule - 1

- A rule or "rule set" is a statement that tells browsers how to render particular elements on an HTML page.
- A rule set consists of a **selector** followed by a **declaration** block.
- Each **declaration** consists of a **property** and a **value**.
- A CSS declaration always ends with a semicolon.
- Declaration groups are surrounded by curly brackets.

CSS Rule/ Syntax



CSS Syntax - Selector

- **Selector**: the selector is used to 'select' or 'target' the HTML element that you want to style.
- This is how we tell the web browser to apply a particular style to a particular html element.
- For instance, let's say you want to style all the paragraphs on your web pages.
- The letter "p" would be our selector name and it targets (selects) all paragraphs.

CSS Syntax - Declaration

- Declaration Block: is everything between the curly brackets.
- What you see between these curly brackets are referred to as ***Property: Value*** pairs.
- This is how we define the formatting properties we want to apply to our web pages.

CSS Rule – CSS Selectors

- Rules or Style Declarations begin with **CSS Selectors.**
- The Selector is very important because it tells the web browser which **HTML** element to style.
- The following are the Core CSS Selectors:
  - Tag or Element
  - Descendant
  - Pseudo-Class
  - Universal
  - Id
  - Class

Tag or Element Selectors

- Tag or element selector is simply the HTML tag you are styling, but without the angular brackets.
- It is simply using the tag name or element name as the selector.
- In most CSS implementations, the tag selector is combined with other selectors
- Tag selectors can be applied as:

body {margin: 0; padding: 0;}

Descendant Selectors

- A descendant selector matches any element that is a descendant of another—in other words, an element that is nested at any level inside another.
- To create a descendant selector, separate the two selectors by a space, and putting the descendant (nested) element on the right.
- For example, the following style rule changes the font size of paragraphs nested at any level inside a **<blockquote>** tag:

blockquote p { font-size: 13px; }

Pseudo-class Selectors

- Pseudo-classes and pseudo-elements are used to apply styles to elements based on their state or position in the document.
- Apply a pseudo-class or pseudo-element by appending it to a basic selector.
- For example, **a:link** applies the :link pseudo-class to the <a> tags.
- The **p:first-letter** applies the :first-letter pseudo-element to the <p> tags.

Universal Selectors

- Universal Selector is indicated with an asterisk *.
- The purpose of this selector is to make style one time and it will be applied to every single elements on the web page
- The most common use of the universal selector is to remove, or zero out, all margins and padding. For example,

*** {margin: 0; padding: 0;}**

Id Selector

- An Id Selector is indicated by preceding its name by the pound # sign.
- Id selectors are used for page elements that appear one time on a web page and require unique formatting instructions.
- For example, an Id selector could be :

#dare { text-indent: 3em }

- This would be referenced in HTML by the **id** attribute as shown here:

** Text indented 3em **

Class Selector

- A class selector's name is preceded by a full stop (.).
- Unlike Id selectors, a class selector can be applied to several elements on a web page.
- An example would be if you want to give all the paragraphs on your web page the same styling.

```
.note {background-color: #013370;
        color: white;
       font-size: small;
}
```

- This would be referenced in HTML by the class attribute of the element:

** This my paragraph **

Styling Background

- CSS background properties are used to define the background effects of an element.
- CSS properties used for background effects:
  - background-color
  - background-image
  - background-repeat
  - background-attachment
  - background-position

Background Color

- The background-color property specifies the background color of an element.
- The background color of a page is defined in the body selector:

body {background-color:#b0c4de;}

- The background color can be specified by:
  - **Name** - a color name, like "red"
  - **RGB** - an RGB value, like "rgb(255,0,0)"
  - **Hex** - a hex value, like "#ff0000“

Background Image

- The background-image property specifies an image to use as the background of an element.
- By default, the image is repeated so it covers the entire element.
- The background image for a page can be set like this:
- **Example**

body {background-image: url('path to the image file');}

Styling Links - 1

- Links can be style with any CSS property such as color, font-family, and background-color.
- Links can be styled differently depending on what state they are in.
- The four link states and the order of styling them are:

  a:link - normal, unvisited link

  a:visited – link's state when the user has visited it

  a:hover – link's state when the user mouses over it

  a:active – link's state the moment it is clicked

Styling Links - 2

- **a:link {color:#FF0000;}** /* unvisited link */
  a:visited {color:#00FF00;} /* visited link */
  a:hover {color:#FF00FF;} /* mouse over link */
  a:active {color:#0000FF;} /* selected link */
- When setting the style for several link states, these order rules must be applied:
  - **a:hover MUST come after a:link and a:visited**
  - **a:active MUST come after a:hover**

Summary -- 1

- CSS gives power to web developers to improve the aesthetics, and layout of web pages .
- External style sheets are recommended for medium to large scale web development projects.
- A CSS rule set consists of a selector and declarations terminated by semi colons.
- The declarations made of **property: value;** are enclosed in curly braces to form a block

Summary -- 2

- **Id** and **class** selectors give you power to create your own selector names.
- When used properly, CSS can transform a raw html structured page into an aesthetically appealing web page.
- The best way to master these skills is to sit up and practice.

Focus Questions

Visit https://www.w3schools.com and read on the following:

1. The CSS Box Model.
2. Layout Design using CSS – CSS Height, Width and Positions
3. Styling Forms
4. Styling Tables

Kindly bring any questions you may have to class for discussion.

Our Next Lecture …

1. The CSS Box Model
2. Styling Forms and Tables in CSS.
3. CSS Positions
4. CSS Specificity